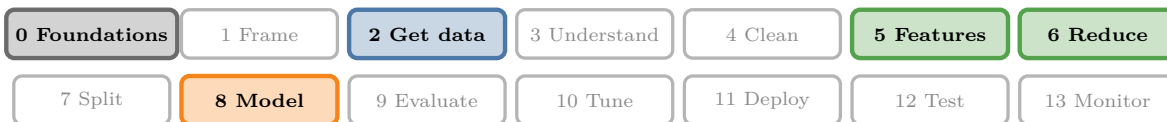


Types of Machine Learning

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Where this fits: Pipeline map steps 0 (Foundations), 2 (Get data), 5 (Engineer features), 6 (Reduce dimensions), 8 (Model). See the [Course map](#).



- **Builds on:** Machine learning ([Note 2](#)).
- **Leads to:** K-nearest neighbours ([Note 6](#)); Applications of ML ([Note 8](#)); Framing an ML problem ([Note 9](#)); Feature selection ([Note 9](#)); Logistic regression ([Note 13](#)); Feature extraction ([Note 23](#)).
- **Compare with:** Feature extraction ([Note 23](#)); Low-rank approximation (truncated SVD) ([Note 612](#)).

1. Overview

Key point: Based on how much supervision they need, ML algorithms fall into four types: supervised, unsupervised, semi-supervised and reinforcement learning.

ML algorithms can be grouped in three different ways:

- by **how much supervision** they need (this Note),
- by whether they learn all at once or keep learning over time (batch vs online),
- by how they generalise to new data (instance-based vs model-based).

Supervision (G-1920) means correct answers that guide the algorithm while it learns. Figure 1 shows the four types this produces and their sub-types.

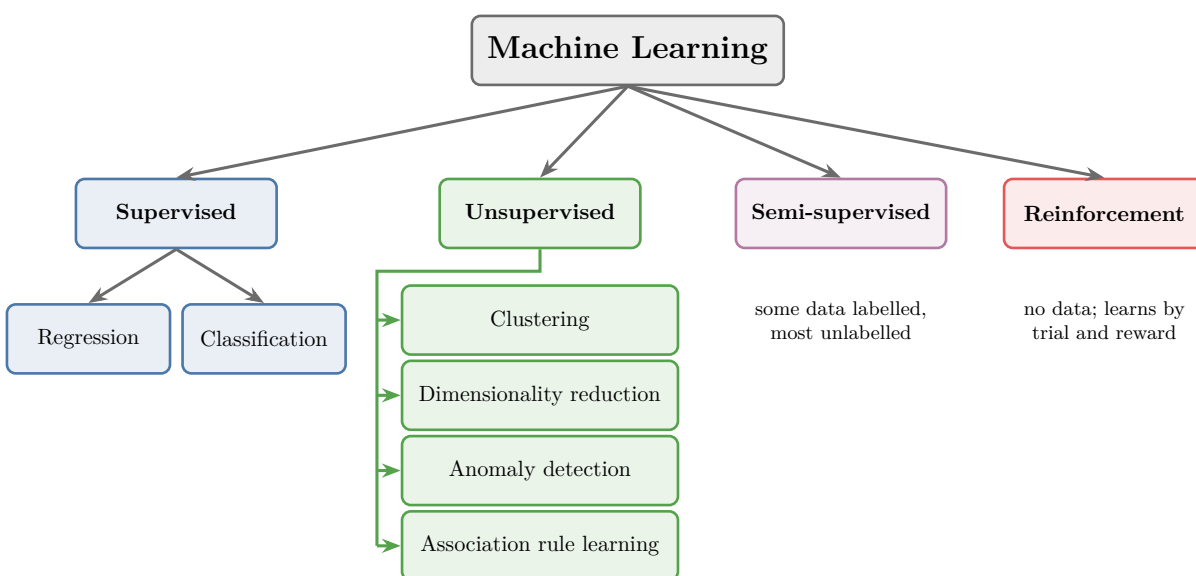


Figure 1: ML types by amount of supervision

2. Supervised learning

Key point: In supervised learning, the data has both inputs and outputs. The algorithm learns how the output depends on the inputs, then predicts the output for new inputs.

2.1 Learning from inputs and outputs

Key point: Features + target $\rightarrow$ learn the relationship $\rightarrow$ predict the target for new observations.

Three words describe any data table:

- a **feature** (G-772) is an input variable (one column of the table);
- the **target** (G-1949) is the output we want to predict (also a column);
- an **observation** (G-1374) is one record (one row of the table).

In **supervised learning** (G-1919), every observation has both its feature values and the correct target value. The algorithm learns the mathematical relationship between them.

Example: campus placements. We have data on 5,000 past students (Figure 2):

- **Features:** IQ and CGPA.
- **Target:** whether the student was placed.
- **Observations:** the 5,000 students, one per row.

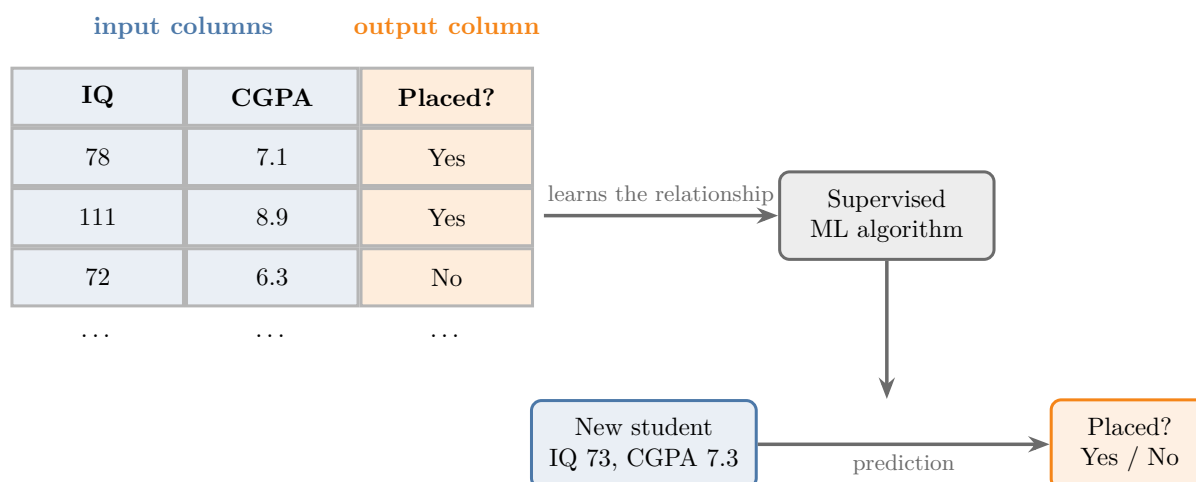


Figure 2: Supervised learning: learn from labelled observations, predict for a new one

After learning, the algorithm can take a new student, say IQ 73 and CGPA 7.3, and predict *yes* or *no*. Most of the ML used in industry is supervised learning.

Extra: The target is also called the **label** (G-1032) or the **output**. Data that includes the target is called **labelled data** (G-1034).

2.2 Numerical and categorical data

Key point: Data is either numerical (numbers) or categorical (categories).

Before splitting supervised learning further, we need the two basic types of data:

Type	What it holds	Examples
Numerical	Numbers	Age, weight, height, IQ
Categorical	Categories	Gender, nationality, phone brand

2.3 Regression and classification

Key point: Numerical output: regression. Categorical output: classification.

Supervised learning has two sub-types. Which one we have depends only on the **target**:

- **Regression** (G-1655): the target is numerical (**numerical data**, G-1367). *Example:* predicting a student's salary package (4.5 LPA, 3.0 LPA, ...) from IQ and CGPA.
- **Classification** (G-395): the target is categorical (**categorical data**, G-351). *Example:* predicting placed / not placed from IQ and CGPA.

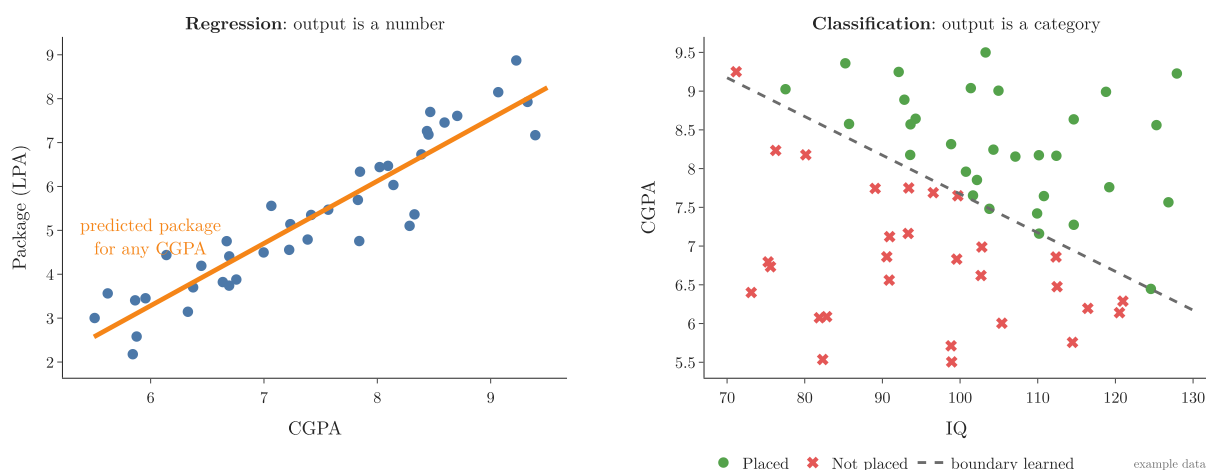


Figure 3: Regression vs classification

In Figure 3, regression fits a line that gives a number for any input. Classification learns a **decision boundary** (G-555): the line or curve that separates the inputs assigned to one category from those assigned to another.

Problem	Output	Type
Price of a house	A number	Regression

Problem	Output	Type
Is this email spam?	Yes / No	Classification
Will it rain today?	Yes / No	Classification
Is there a dog in this image?	Yes / No	Classification

3. Unsupervised learning

Key point: In unsupervised learning, the data has features only. There is no target to predict, so the algorithm finds structure instead: groups, fewer features, oddities or links.

3.1 Learning from inputs only

Key point: No target means no prediction. The algorithm describes the data instead.

In **unsupervised learning** (G-2058), the data has only features. For example, we have the IQ and CGPA of students, but no placement target. Figure 4 shows the same table as Figure 2 with the target column removed.

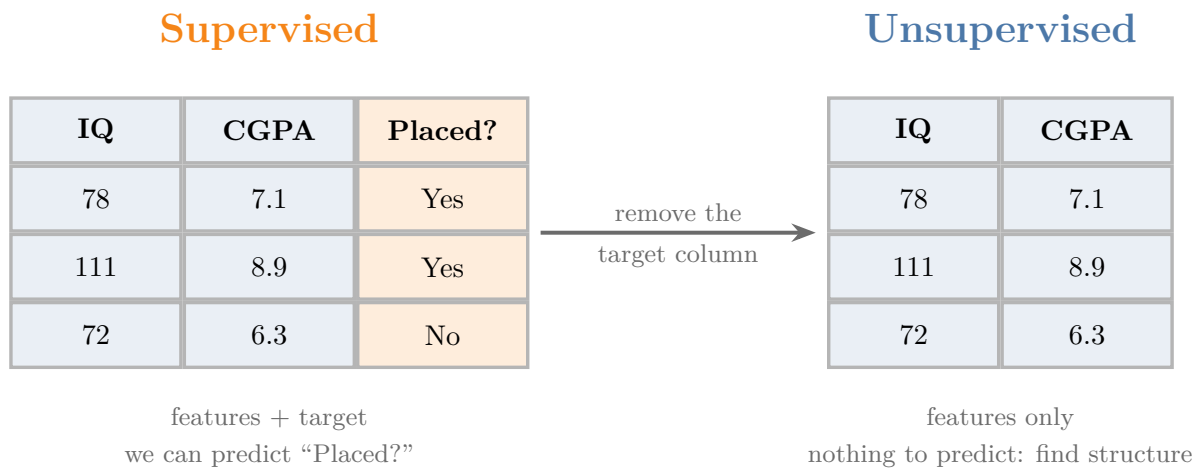


Figure 4: The same table with and without the target column

Figure 5 shows the same change on a scatter plot of 90 example students.

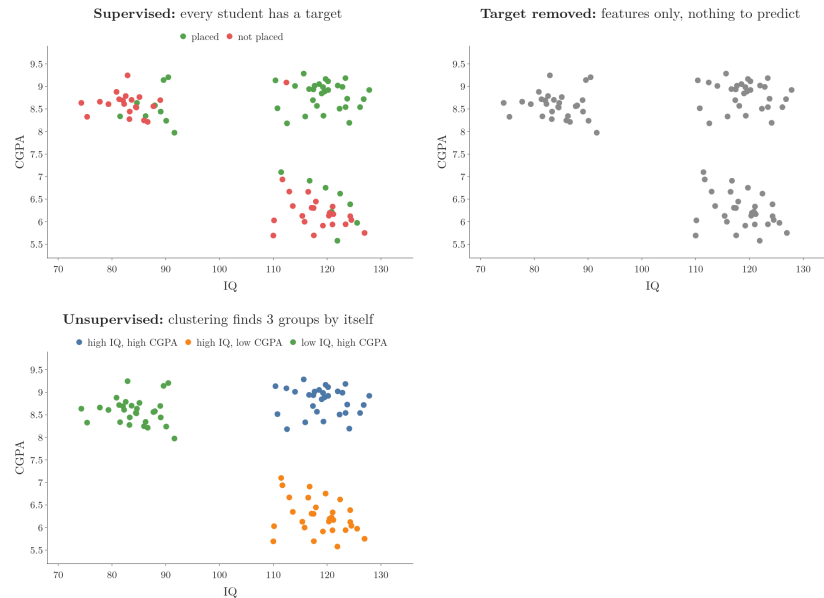


Figure 5: One dataset, seen three ways: with a target (supervised), with the target removed, and after clustering (example data)

1. **Supervised.** Every dot has a colour: placed or not placed. The colour is the target, so an algorithm can learn to predict it.
2. **Target removed.** Every dot is grey. Only IQ and CGPA are left, and there is nothing to predict.
3. **Unsupervised.** A clustering algorithm (Section 3.2) looks only at IQ and CGPA and colours the three groups it finds. The groups are not “placed” and “not placed”: the algorithm never saw that column.

Without a target, we cannot predict anything. Unsupervised learning performs four other jobs:

- clustering (Section 3.2);
- dimensionality reduction (Section 3.3);
- anomaly detection (Section 3.4);
- association rule learning (Section 3.5).

3.2 Clustering

Key point: Clustering finds groups of similar observations, without being told what the groups are.

Clustering (G-401) splits the data into groups (**clusters**, G-399) of similar observations. Figure 6 shows students plotted by IQ and CGPA; the algorithm found three groups on its own.

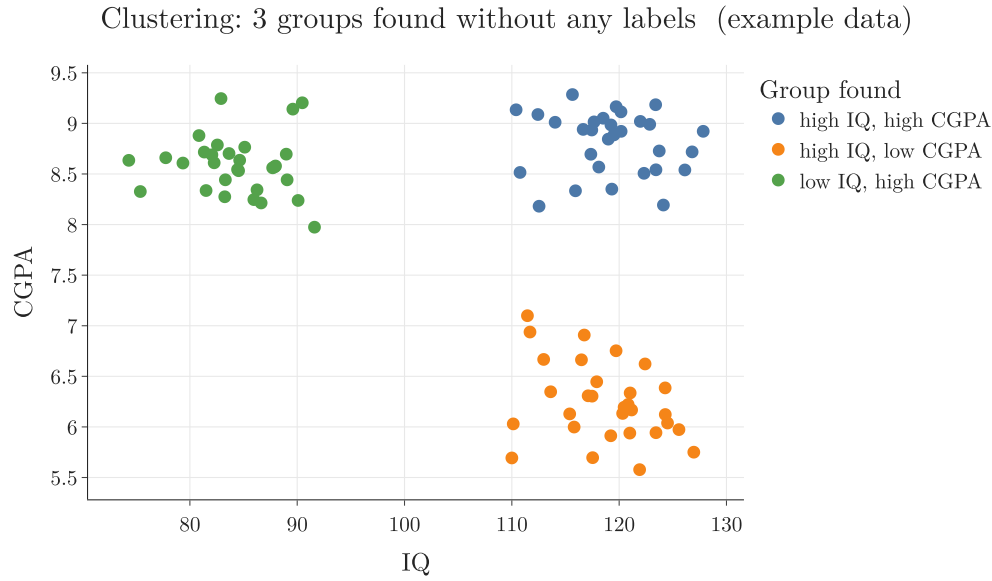


Figure 6: Clustering: groups found without labels

What the groups give us:

- **A category for each student:** for example, high IQ but low CGPA. A new student can be placed in one of the groups.
- **Labels for free:** we can number the groups (0, 1, 2) and then use them as a target for supervised learning.
- **Customer segments:** an e-commerce site can group its customers by how they behave and treat each group differently.

With two features we could spot the groups by eye. Clustering also works with hundreds of features, where no human can see the groups.

3.3 Dimensionality reduction

Key point: Dimensionality reduction cuts down the number of features while keeping the information. Fewer features speed up learning and let us plot high-dimensional data.

Each feature is a **dimension** (G-610). (A tensor's dimensions mean something else, its number of axes: see Section 5 of the [tensors Note](#).) Images and text can have thousands of features, which causes two problems:

1. Algorithms become **slow**.
2. After a point, extra features **stop improving** the results.

Dimensionality reduction (G-611) removes the extra features.

Example: house prices. Number of rooms and number of washrooms carry related information. We can combine them into one feature, area (Figure 7).

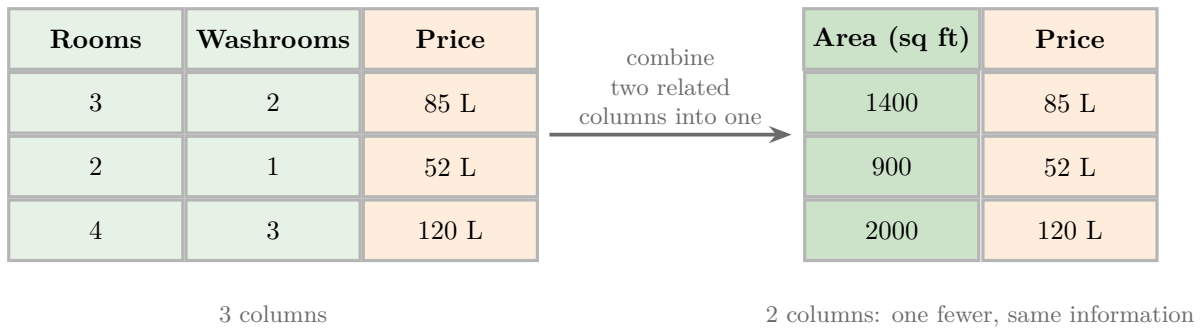


Figure 7: Combining two related features into one

The data has one feature fewer and loses almost no information. Making such a feature by hand, from domain knowledge, is called **feature construction** (G-760; see Section 7 of the [feature engineering Note](#)).

When an algorithm such as PCA computes the new features from the data instead, with no domain knowledge, the process is called **feature extraction** (G-762; see Section 9 of the [feature engineering Note](#)).

Visualisation. A graph can show at most 3 dimensions. To see data with hundreds of features, we reduce them to 2 or 3 features and plot those.

Example: handwritten digits. Each digit is an 8 x 8 grid of pixels, which makes 64 features, one per pixel. Figure 8 reduces them to 3 features: images of the same digit land close together.

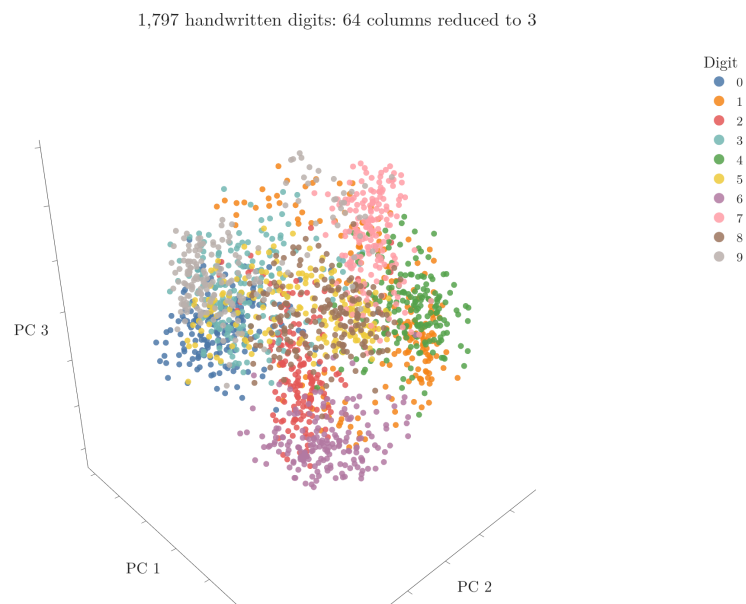


Figure 8: Handwritten digits: 64 features reduced to 3

The technique used for Figure 8 is **PCA** (principal component analysis, G-1562), covered in detail in later Notes. The Notebook for this Note (`notebook.ipynb`) shows Figure 8 as a 3D plot that we can rotate.

Extra: The best-known digits dataset, MNIST, uses 28 x 28 pixel images, which gives 784 features (LeCun et al. 1998). The idea is the same.

3.4 Anomaly detection

Key point: Anomaly detection learns what normal data looks like and flags anything far from it.

Anomaly detection (G-201) finds observations that do not fit the pattern of the rest. Typical uses:

- spotting defects in manufacturing,
- catching credit card fraud,
- flagging unusual loan applications.

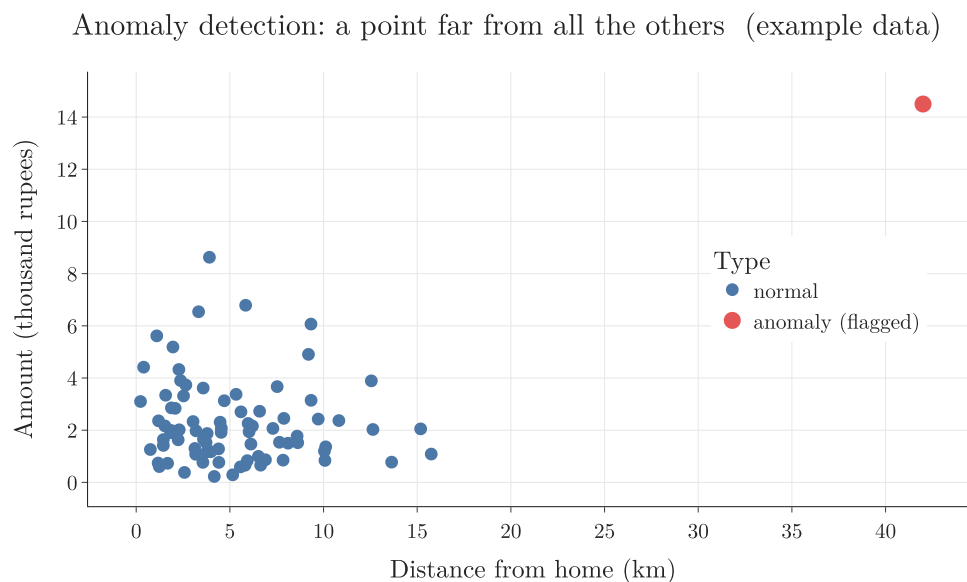


Figure 9: Anomaly detection: one point far from the rest

In Figure 9, almost all transactions are small and close to home. The red one, large and 42 km away, is far from every normal point, so the system flags it.

3.5 Association rule learning

Key point: Association rule learning finds items that tend to occur together.

Association rule learning (G-218) looks for “if this, then that” patterns in data. Its classic use is deciding which products a shop places next to each other.

Example: a supermarket. We scan all the bills from the last one or two years (Figure 10). Suppose milk appears in 8 of 100 bills, and eggs appear in 6 of those 8. Then people who buy milk usually buy eggs as well, so the shop places the two together.

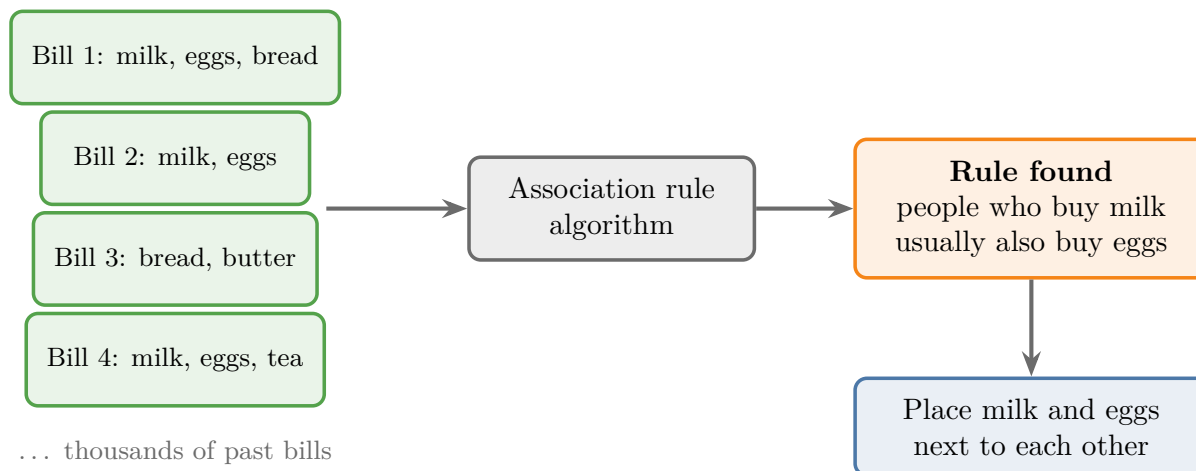


Figure 10: Association rules: from past bills to shelf placement

A famous case: an analysis of shopping baskets at a US store chain found that customers buying baby diapers in the evening often also bought beer. Hidden patterns like this are hard to spot by hand and easy for an algorithm.

Extra: The diapers-and-beer story is usually told about Walmart, but it traces back to a 1992 study of shopping baskets at Osco Drug stores, which did find beer and diapers bought together in the evening. The shops never moved the two products together, so the popular claim that shelf placement boosted sales is a legend (Power 2002). The beer-and-diapers find is still the standard example of association rules.

4. Semi-supervised learning

Key point: Semi-supervised learning labels a few observations by hand and lets the algorithm label the rest.

Labels are **expensive**. Collecting inputs is easy; for example, we can download thousands of images in minutes. But someone has to look at each image and write down what is in it, which costs time and money.

Semi-supervised learning (G-1768) works with data where only a small part is labelled. We label a few observations, and the algorithm labels the rest automatically.

Example: Google Photos (Figure 11).

1. The app groups photos by the faces in them, without knowing any names. This step is unsupervised.

2. We tag one photo with a name, such as “Dad”.
3. The name spreads to every photo in that group.

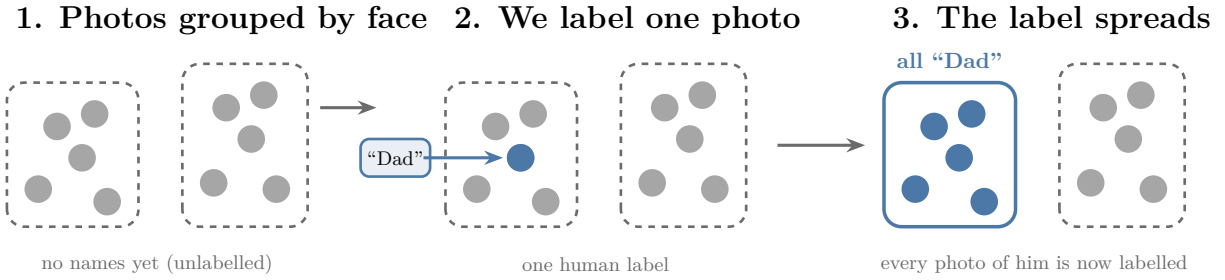


Figure 11: Semi-supervised learning: one label spreads to a whole group

One human label replaced hundreds.

How does one label reach a whole group? Similar observations sit close together, so a label can hop from each observation to its **nearest neighbours** (G-1306), the observations closest to it. Figure 12 shows the hops on the 90 students of Figure 6.

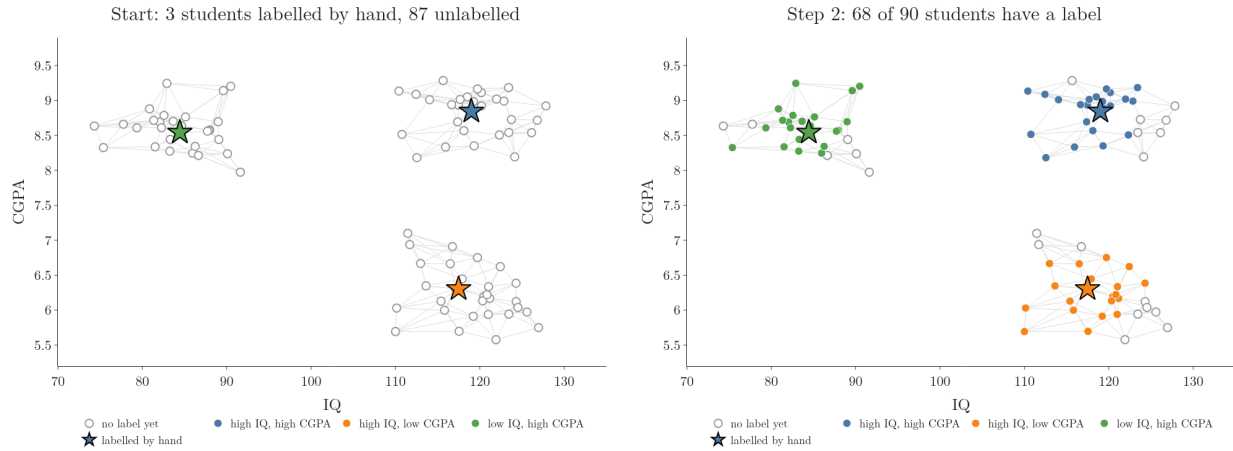


Figure 12: Label propagation on the 90 students of Figure 6. Stars: the three students labelled by hand. Grey links join each student to its 5 nearest neighbours; a white dot has no label yet.

1. **Start.** We label one student in each group by hand (the stars). The other 87 have no label.
2. **Each step.** Every student looks at its linked neighbours, adds up the labels they carry, and takes the strongest one. The three hand labels never change.
3. **The spread.** After step 1, 24 of the 90 students have a label; after step 2, 68; after step 3, 89; after step 4, all 90.
4. **The result.** Every student ends up with the label of its own group, although only three were labelled by hand.

This method is called **label propagation** (Zhu and Ghahramani 2002). It works only when observations with the same label sit close together, as the three groups do here.

5. Reinforcement learning

Key point: In reinforcement learning, there is no ready-made dataset. An agent collects its own experience by acting, gets rewards or punishments, and improves its rules.

5.1 Agent, environment, policy and reward

Key point: The agent acts, the environment rewards or punishes, and the agent updates its policy. Repeat.

Reinforcement learning (RL) (G-1660) starts with no ready-made dataset: nobody hands the algorithm a table of examples with correct answers. The algorithm produces its own data by acting. Each action, and the reward or punishment that follows it, is one piece of experience, and the algorithm learns from this experience, getting better over time (Sutton and Barto 2018, §1.1).

The parts of an RL system:

- **Agent** (G-179): the learner, for example a self-driving car or a game-playing program.
- **Environment** (G-694): the world the agent acts in, for example the road or the game board.
- **Policy** (G-1512): the agent's rule book, saying which action to take in each situation.
- **Reward / punishment** (G-1690): feedback from the environment after each action, good or bad.

Figure 13 shows the loop:

1. The agent observes the environment.
2. Following its policy, the agent takes an action.
3. The environment returns a reward or punishment.
4. The agent updates its policy.

The goal is to collect as much reward and as little punishment as possible.

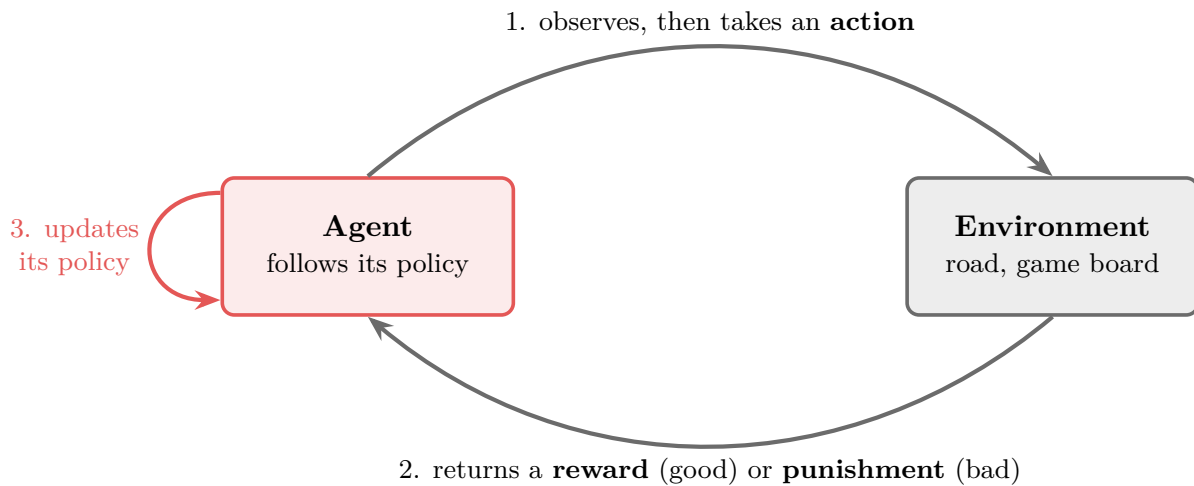


Figure 13: The reinforcement learning loop

5.2 Example: fire or water

Key point: A bad result changes the policy, so the next choice is better.

Figure 14 shows an agent that can walk to fire or to water.

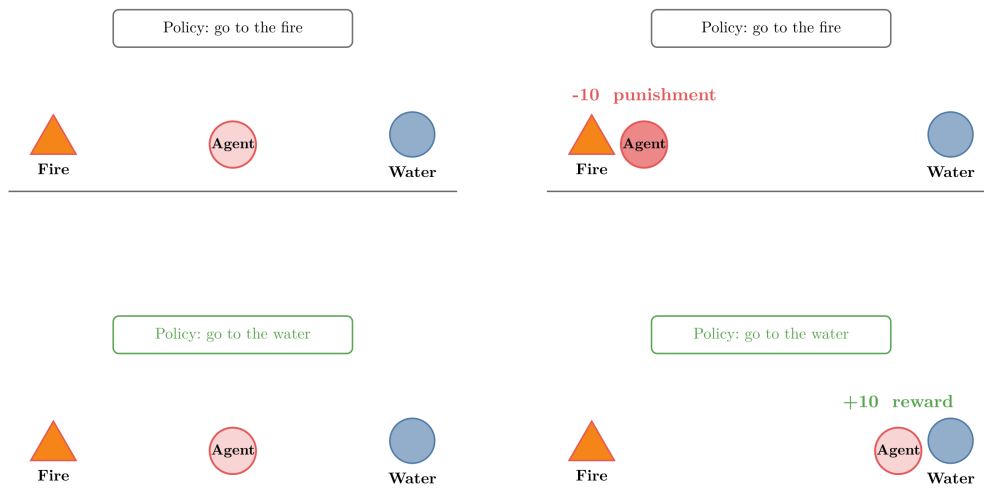


Figure 14: An agent learning to choose water over fire

1. The agent's policy says *go to the fire*. The agent goes, and gets a punishment.
2. The agent updates its policy to *go to the water*.
3. The agent goes to the water and gets a reward.

People learn many things the same way: nobody hands us a dataset for living in a new city; we try, make mistakes and adjust. Training a pet with treats works the same way.

5.3 Where RL is used

Key point: RL beat the world champion at Go and is used for self-driving cars, but it is harder to build than the other types.

- **Self-driving cars:** the car learns to drive by acting on the road and receiving feedback.
- **Games:** Go is far more complex than chess, and beating top humans was thought to be years away. In March 2016, DeepMind's agent **AlphaGo** beat world champion Lee Sedol in four out of five games (DeepMind 2016).

RL is harder to set up than the other types, but its use is growing fast.

6. Summary

Type	Data	Goal	Example
Supervised: regression	Inputs + numerical output	Predict a number	Salary package from IQ and CGPA
Supervised: classification	Inputs + categorical output	Predict a category	Placed or not
Unsupervised: clustering	Inputs only	Find groups	Customer segments
Unsupervised: dimensionality reduction	Inputs only	Fewer features	Rooms + washrooms → area
Unsupervised: anomaly detection	Inputs only	Flag unusual observations	Card fraud
Unsupervised: association rules	Inputs only	Find items that go together	Milk and eggs
Semi-supervised	A few labels, many unlabelled observations	Label the rest	Google Photos faces
Reinforcement	No ready-made data; experience and rewards collected from an environment	Learn the best actions	AlphaGo

- To identify a supervised problem, look at the **target**: number → regression, category → classification.
- No target → unsupervised.
- Few labels → semi-supervised.
- No ready-made data, only feedback on the agent's own actions → reinforcement.

7. Sources

Built from

- CampusX, “Types of Machine Learning for Beginners | Types of Machine learning in Hindi | Types of ML in Depth”, YouTube, <https://www.youtube.com/watch?v=81ymPYEtFOw>

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- Zhu, X. and Ghahramani, Z. (2002). Learning from Labeled and Unlabeled Data with Label Propagation. Technical Report CMU-CALD-02-107, Carnegie Mellon University.

8. Key terms

Term	Meaning
Supervision	Correct answers that guide an algorithm while it learns
Supervised learning	Learning from data with inputs and outputs, to predict outputs
Feature	An input variable; one column of the data table
Target	The output we want to predict; also called the label
Observation	One record; one row of the data table
Labelled data	Data that includes the target
Numerical data	Data made of numbers
Categorical data	Data made of categories
Regression	Supervised learning with a numerical output
Classification	Supervised learning with a categorical output
Unsupervised learning	Learning from inputs only, to find structure
Clustering	Splitting data into groups of similar observations
Cluster	One group found by clustering
Dimension	One feature
Dimensionality reduction	Reducing the number of features while keeping the information
Feature extraction	New features computed from existing ones by an algorithm such as PCA
PCA	Principal component analysis, a dimensionality reduction technique
Anomaly detection	Finding observations that do not fit the pattern of the rest
Association rule learning	Finding items that tend to occur together
Semi-supervised learning	Learning from a few labelled observations and many unlabelled ones
Label propagation	Semi-supervised method in which labels hop from labelled observations to their nearest neighbours, step by step
Reinforcement learning	Learning by acting and receiving rewards or punishments
Agent	The learner in reinforcement learning
Environment	The world the agent acts in
Policy	The agent's rules for which action to take

Term	Meaning
Reward / punishment	Good / bad feedback after an action